

Boukhobza Abderraouf

Computer Science Student | Backend Developer

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PROFESSIONAL PROFILE

Master's student in Artificial Intelligence with experience in backend development using Node.js, NestJS, TypeScript, REST APIs, MySQL, and Docker. Also skilled in Android development with Kotlin and Jetpack Compose. Interested in software architecture, cloud technologies, and AI-powered applications, with strong analytical and problem-solving abilities.

EDUCATION

Master's Degree in Artificial Intelligence | Mohammed Seddik Benyahia University – Jijel, Algeria
2025– Present

Bachelor's Degree in Computer Science | Mohammed Seddik Benyahia University – Jijel, Algeria
2022 – 2025

PROJECTS

AI-Powered Grading & Exam Management System | Academic Project

- Developed a NestJS backend utilizing local LLM models to automate student answer sheet scoring and qualitative feedback generation.
- Engineered an asynchronous job queue using BullMQ and Redis to isolate heavy LLM workloads from the main API thread, improving server responsiveness.
- Established real-time bidirectional communication using WebSockets to notify users of grading milestones.
- Implemented secure, role-based student and teacher workflows with JWT authentication, SQLite, and TypeORM.

Technologies: NestJS, TypeScript, Ollama, BullMQ, Redis, WebSockets (Socket.io), TypeORM, SQLite, JWT

E-Commerce Backend Infrastructure

- Developed a REST API using NestJS (TypeScript), TypeORM, and MySQL for e-commerce order workflows.
- Prevented checkout race conditions and double-selling using transactional operations and pessimistic row-locking.
- Integrated a Redis caching layer with custom invalidation triggers to reduce database read latency.
- Programmed an automated order synchronization service with custom cell-formatting using the Google Sheets API.
- Secured the platform using rate limiting (NestJS Throttler), RBAC guards, and structured logging with Pino.
- Containerized the application stack (API, MySQL, Redis) using Docker and Docker Compose

Technologies: NestJS, TypeScript, MySQL, TypeORM, Redis, Docker, Google Sheets API, JWT, Pino.

Restaurant Ordering Backend | Academic Project [github-link](#)

- Built a REST API with Laravel (PHP) and Eloquent to power restaurant ordering: order creation/update/show/cancel, OrderDetails and Payment lifecycle, and table-status management.
- Enforced business rules with comprehensive request validation, Eloquent relationships, and centralized controller logic for reliable workflows.

- Improved developer experience and observability using Scribe-generated API docs and Laravel Telescope for request/query/exception tracing.

TECHNICAL SKILLS

Languages: TypeScript, PHP, Python , java

Backend Development: NestJS, Laravel, REST APIs, Authentication & Authorization (JWT)

Mobile Development: Android Development, Jetpack Compose, Kotlin Coroutines, HTTP Clients

Databases: SQL , MySQL , SQLite , MongoDB , Redis

Caching & Messaging: Redis, Caching Strategies, Queue Processing , BullMQ

Tools & Platforms: Git, GitHub, Postman

Software Architecture: Clean Architecture, SOLID Principles, MVVM, Repository Pattern

SOFT SKILLS

- Problem Solving
- Teamwork & Collaboration
- Communication
- Time Management
- Adaptability
- Attention to Detail

LANGUAGES

Arabic: Native / Fluent

English: B1

French: A2